

VARIKUPPALA RAVINDER

Senior AI Platform Engineer

Agentic AI | Multi-Agent Systems | RAG | AI Platform Architecture | FastAPI | AWS | React | Node.js
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CAREER SUMMARY

Senior AI Platform Engineer with 6+ years of experience designing and delivering enterprise-scale AI platforms, distributed systems, and full-stack applications. Specialized in Agentic AI, Multi-Agent Systems, LangGraph, Retrieval-Augmented Generation (RAG), MCP Tooling, and Cloud-Native Architectures. Proven track record of building production AI systems that reduced manual workflows by 40%, improved retrieval accuracy by 30–40%, and delivered sub-2 second intelligent search experiences. Passionate about architecting scalable, secure, and production-ready AI platforms from backend services to cloud infrastructure.

TECHNICAL SKILLS

Languages: Python • JavaScript • TypeScript

AI Engineering: Agentic AI • Multi-Agent Systems • LangGraph • RAG • MCP • Tool Calling • Prompt Engineering • Embeddings • Vector Search • Semantic Retrieval • AI Guardrails

Backend: FastAPI • Node.js • Express.js • REST APIs • Server-Sent Events (SSE) • WebSockets

Frontend: React.js • Next.js • Redux • Material UI • Tailwind CSS

Databases: PostgreSQL • MySQL • DynamoDB • Redis • Qdrant • pgvector

Cloud & DevOps: AWS (ECS, EC2, Lambda, API Gateway, IAM, CloudFormation) • Docker • GitHub Actions • CI/CD

Architecture: Distributed Systems • Microservices • Event-Driven Architecture • System Design • Planner–Executor Architecture • API Gateway • Streaming Applications

WORK EXPERIENCE

Senior Software Engineer | EPAM Systems

2024- Present

Designed and architected enterprise-scale AI platforms leveraging Agentic AI, Multi-Agent Systems, RAG, and cloud-native technologies to automate business workflows and intelligent knowledge retrieval.

- Architected **LangGraph-based planner-executor multi-agent systems** with supervisor orchestration, enabling intelligent retrieval, workflow automation, and tool execution.
- Designed and optimized **RAG pipelines** over millions of enterprise documents, improving retrieval accuracy by **30–40%** and achieving **sub-2 second** response latency through advanced chunking, metadata enrichment, and vector search.
- Built **MCP tool servers** enabling AI agents to autonomously interact with Jira and enterprise APIs, significantly reducing manual operational effort.
- Developed scalable AI applications using **FastAPI, React, Next.js, and Server-Sent Events (SSE)**, delivering real-time streaming user experiences.
- Established production-ready AI engineering practices including **guardrails, retry strategies, CI/CD pipelines, AWS infrastructure, and architecture standards** to improve platform reliability and scalability.
- Led the development of **Elsa AI Assistant**, a flagship enterprise AI platform integrating RAG, Multi-Agent orchestration, and enterprise workflow automation.

Specialist – Software Engineering | LTIMindtree

2022- 2024

- Architected data-intensive microservices with clear service boundaries, API contracts, and data flow patterns serving enterprise clients at scale
- Built full-stack features across frontend and backend layers, data transformation pipelines improved throughput by 20–30% and reduced end-to-end latency for downstream consumers
- Led system design reviews and drove production deployments across multiple client engagements

Senior Developer | DevRabbit IT Solutions

2019- 2022

- Delivered full-stack applications end-to-end - owning frontend, backend, and data layer across the complete delivery lifecycle for enterprise clients
- Designed RESTful APIs and reusable component libraries reducing codebase complexity and cutting feature delivery time by ~20%
- Introduced automation scripts for build tooling and data processing, establishing a clean polyglot architecture pattern

PROJECTS

AI Career Platform (Personal | Production)

Designed and deployed a production-ready AI-powered career platform demonstrating end-to-end AI platform engineering capabilities.

- Built a **LangGraph-based agentic recruiter assistant** using Planner–Executor architecture with RAG over **Qdrant Cloud**.
- Designed a **PostgreSQL-first architecture** ensuring a single source of truth for profile, projects, and AI knowledge ingestion.
- Developed a complete **Admin CMS** enabling browser-based content management without code deployments.
- Built streaming AI interactions using **FastAPI, Server-Sent Events, and Next.js**.

Deployed the complete platform on **AWS EC2** using **Docker Compose, Redis, Caddy, and AWS RDS**.

Tech Stack: FastAPI • LangGraph • PostgreSQL • Qdrant • Redis • Docker • AWS • Next.js

Elsa AI Assistant — Enterprise AI Platform

Enterprise-grade AI platform supporting intelligent knowledge retrieval and workflow automation.

- Architected **Planner–Executor Multi-Agent Architecture** with supervisor-driven orchestration.
- Built RAG-based enterprise search over millions of documents with streaming AI responses.
- Developed MCP integrations enabling autonomous Jira operations and enterprise tool execution.
- Implemented Kafka-driven asynchronous processing, audit dashboards, RBAC, and production guardrails.

Impact: 35% reduction in Jira ticket creation time · 40% improvement in query resolution accuracy · Sub-2s RAG response latency

Tech Stack: Python • FastAPI • LangGraph • Next.js • Kafka • Qdrant • AWS ECS • Docker

Healthcare & Enterprise Platforms

Fleet Manager (IoT monitoring) • AI FinOps Copilot • Ayurway (EMR system) • iDestination (location intelligence) • Notifii (enterprise package tracking) - full-stack systems with real-time dashboards

ACHIEVEMENTS

Ace of Delivery Award - EPAM • Client Hero Award - EPAM • Fearless Problem Solver Award (3x) - EPAM & LTIMindtree • A-TEAM SPOT ON Award - LTIMindtree • Super Crew Award - LTIMindtree

EDUCATION AND CERTIFICATION

Bachelor of Technology (CSE) Guru Nanak Institutions Technical Campus, JNTU	2019
Diploma (CSE) Govt Polytechnic Masabtank, SBTET	2016